GS EXTERIOR EXPERTS · DECK DIVISION · TAKEOFF + BUILD SET

5549 S Lansing Ct · Elevated Deck · Englewood

5549 S Lansing Ct, Englewood · Rachel Chiu · 8' x 12' deck, 10' above the yard, off the existing door · TrueFrame 2x10 @ 16" on a (2)4x10 Douglas fir beam and (2) 6x6 DF #1 posts on Diamond Piers · TimberTech Reserve Driftwood, single picture frame, CONCEALoc · TimberTech Fulton rail, Black · posts + anchors for the owner-supplied square shade sail

SOLD · CHECK OR ACH	COST	GROSS PROFIT	MARGIN	FINANCED	SQUARE FEET
$13,000	**$8,856**	**$4,144**	**31.9%**	**$13,978**	**92**

DESIGN READS

Frame	7'-9" x 11'-5 1/2" outside to outside
Deck	7'-11 1/8" x 11'-6 9/16" over fascia · 92 SF · 10'-0" above grade
Joists	7 2x10 #1 True Frame Joist SYP GC @ 16" OC x 10'-9"
Rims	2-ply 2x10 sides; front rim = flush beam
Beam	FRONT FLUSH (2)4x10 DF#1: 7'-9", 2 posts at 9" / 7'-3" (worst 1,794 lb)
Posts	2 x 6x6 Douglas fir #1 with 4x4 knee braces on DP-50/50
Load	50 psf (40 psf live governs) · engineered
Decking	TimberTech Reserve Sand — 24 rows parallel to the house, picture frame in Reserve Driftwood, 3/16" gaps, CONCEALoc
Rail	Fulton 36" Black: 5 bays, 6 posts, 31 LF

- nominal 8'-0" x 12'-0" -> frame 7'-9" x 11'-5 1/2": field boards cut to 7'-0", 24 full rows, deck over fascia 7'-11 1/8" x 11'-6 9/16"
- front flush beam IS the front rim — joists hang into it, posts under it, no separate rim plies

SITE & CODE FLAGS internal

ENGINEER ENGINEERING FRONT FLUSH (2)4x10 DF#1: (2)4x10 not in IRC R507.5 — treated as (3)2x10; engineer to confirm IRC Table R507.5

ENGINEER ENGINEERING Shade sail: the sail posts stand on the deck as the rail corners — base-plated on top of the decking over solid blocking: the sail's wind pull goes through the base plate bolts into the blocking, the frame, the knee-braced 6x6 posts and the house ledger:

		engineer checks the post-to-rim bolting, the frame and the lateral system for it ASCE 7 / IRC R507
ENGINEER	ENGINEERING	Shade sail: sail wind load (115 mph) on the 2 steel posts and the house anchors — engineer sizes the posts / footings and the anchor connection into the house framing (not into siding or fascia alone) ASCE 7 / IRC R301.2
ENGINEER	LEDGER	house floor cantilevers past the foundation — ledger attachment to a cantilevered floor needs an engineered detail or a freestanding deck IRC R507.9.1
ENGINEER	LEDGER	Arapahoe County deck guide: a ledger 'shall not be attached to a cantilevered floor' — attachment to a cantilever requires engineer approval. The 2' second-floor cantilever runs the full rear wall (assessor sketch): the stamped design either details the ledger into the cantilevered joists / blocking with the lateral ties back to the floor system, or makes the deck freestanding (a second beam on two posts at the house) AC-amended ICC Deck Guide (2021 IRC) / IRC R507.9.1
CODE	FOOTING	county guide: pier footings 12" x 36" minimum unless engineered — submit the Diamond Pier DP-50 evaluation report as the engineered alternate (frost 36") and have the engineer list it on the stamped set Arapahoe deck permit guide (rev. 8/20/2024) / IRC R403.1.4
CODE	RAIL	guards above 30": 36" min., openings under 4", 200 lb point load + 50 psf infill — submit the TimberTech Fulton rail code report / ESR IRC R312 / Table R301.5
CODE	SUBMITTAL	county submittal: site plan with lot / block and distances to every line, scaled drawings with every span, member, hanger, beam build-up and post anchorage, ledger fastener type / size / spacing, the completed county-amended ICC deck guide (page 3), and the engineer's stamp (outside the IRC tables: cantilever attachment, 2-post (2)4x10) Arapahoe Residential Deck Permit guide (rev. 8/20/2024)
CHECK	SETBACK	easement on record: Check plat R1535678 for rear / side utility and drainage easements — no footings or structure inside it plat
CHECK	SITE	shade sail: check whether the county wants a separate permit for the sail structure; the sail must meet the setbacks county
CHECK	FOOTING	Diamond Pier: pins 50" — locate utilities (811) before driving; bearing 1,794 lb/post vs 3,300 lb allowable (manufacturer chart, confirm)
CHECK	LEDGER	verify the rim board behind the ledger is solid 2x/LVL/engineered rim (not I-joist web or a hollow band) before the ledger fasteners go in IRC R507.9.1.1
CHECK	SETBACK	R-P PUD Cherry Creek Vista: rear setback 20' for single-family detached, 10' under the Units 6 & 13 standard (the lot is 6,401 SF, under the 7,000 SF SFD minimum). GIS puts the new deck ~13' off the west line — passes at 10', fails at 20'. Decks get no projection allowance (LDC). Zoning Division to confirm the column and which line is the rear (corner lot) before submittal; pull P75-028 and plat R1535678 2nd Amended PDP P75-027 (1975) / Arapahoe LDC

CHECK	SETBACK	corner lot: E Berry Ave street side needs 10' — deck ~23'-8" clear; south interior side 5' — deck ~33' clear PDP P75-027
CHECK	SITE	HOA: Cherry Creek Vista II HOA (PO Box 3361, Englewood 80155, 303-770-1469) — confirm whether architectural approval is needed county GIS
CHECK	SITE	811 locate before the piers · FEMA zone X · not in the county's draft wildfire-resiliency area (no WUI requirements) · South Metro Fire Rescue Colorado 811 / FEMA / Arapahoe GIS
INFO	SETBACK	HOA — architectural approval before start (color, rail, height)
INFO	ENGINEERING	posts 8'-9 5/16": 4x4 knee braces both directions at every post are in the takeoff — engineer confirms the lateral design DCA6 / IRC R507.4
INFO	LEDGER	lateral: 4 x DTT1Z (750 lb each) into house floor framing IRC R507.9.2
INFO	WIND	basic wind speed 115 mph — standard connectors
INFO	SITE	keep any basement egress window under or near the deck clear (county guide) Arapahoe deck permit guide

THE ORDER Decks & Docks · GSEXT1 · ship to the job

Lumber

NET	ORDER	UNIT	ITEM	WHY
13	**14**	ea	2x10x12 #1 True Frame Joist SYP GC	+1 cull (5 field joist · 2 PF joist · 2 left rim ply · 2 right rim ply · blocking, ledger · blocking, knee-brace block)
2	**2**	ea	4x10x8 Douglas Fir #1	exact (2 flush beam: FRONT FLUSH (2)4x10 DF#1)
2	**2**	ea	6x6x10 Douglas Fir #1	exact (yields 2 posts at ~8'-9 5/16")
2	**3**	ea	4x4x8 Douglas Fir #1 — knee braces	+1 (4 braces at 3'-9 15/16", 45° plumb cuts, 2'-6" legs: one along the beam, one up to the joists at every post)

Footings

NET	ORDER	UNIT	ITEM	WHY
2	**2**	ea	Diamond Pier DP-50/50 (head + 4 x 50" pins, plugs, caps)	exact — confirm stock
2	**2**	ea	Simpson ABA66Z adjustable post base 6x6 on the pier bolt	exact

Hardware

NET	ORDER	UNIT	ITEM	WHY
4	**5**	ea	1/2" x 10" HDG hex bolt + nut + 2 washers — knee braces (brace to the 6x6 post)	+1
2	**3**	ea	1/2" x 12" HDG hex bolt + nut + 2 washers — knee braces (brace to the (2)4x beam)	+1

NET	ORDER	UNIT	ITEM	WHY
2	**3**	ea	1/2" x 6" HDG hex bolt + nut + 2 washers — knee braces (brace to the joist block)	+1
14	**14**	ea	Simpson LUS28Z face-mount hanger 2x8/2x10	exact (7 joists x 2 ends)
4	**4**	ea	Simpson HUCQ210-2-SDS concealed-flange double 2x10 hanger (SDS screws included)	exact (2 side rims x 2 ends) — confirm stock
1	**1**	box	Camo 9ga HDG hanger nail 3" 5 lb (~350)	84 needed of ~350
1	**1**	box	Camo 9ga HDG hanger nail 1-1/2" 5 lb (~650)	56 needed of ~650
1	**1**	box	FastenMaster LedgerLOK 3-5/8" 50 ct	9 needed
4	**4**	ea	Simpson DTT1Z lateral tie kit w/ SDS screws	exact (2 near each end of every ledger, into house floor framing)
2	**2**	ea	Simpson CCQ column cap, 7" seat for the (2)4x10 beam on a 6x6 post (SDS screws incl.) — confirm SKU / engineer's equal	exact
10	**12**	ea	1/2" x 8" HDG carriage bolt + nut + washer	+2 ((2)4x beam plies through-bolted, 2 rows staggered @ 24" OC, 2" from top and bottom)
1	**1**	box	Simpson SD10212R100 #10 x 2-1/2" connector screw 100 ct	24 needed (bases 24 · caps 0)
1	**1**	box	GRK RSS #10 x 3-1/8" structural screw 100 ct	72 needed (laminations 48 · blocking 24)
12	**14**	ea	7/16" x 4-1/2" HDG hex bolt + nut + 2 washers	+2 (6 rail posts x 2)

Flashing & waterproofing

NET	ORDER	UNIT	ITEM	WHY
1	**1**	set	Ledger flashing set: SA membrane 6"x75' + (2) 10' alum. Z + end dams	exact
2	**2**	roll	Nichigo G-Tape 3035 2" x 65' black	90' needed of 130'
1	**1**	roll	Nichigo G-Tape 3035 4" x 65' black	29' needed of 65'

Decking

NET	ORDER	UNIT	ITEM	WHY
12	**14**	ea	TimberTech Reserve Driftwood 1x6x16 Grooved	+2 (A: 24 rows @ 7'-0", 2 per 16' board)
1	**1**	ea	TimberTech Reserve Driftwood 1x6x12 Square Edge — borders	exact (right border 11'-6 9/16")
1	**2**	ea	TimberTech Reserve Driftwood 1x6x20 Square Edge — borders	+1 (left border 11'-6 9/16" + front border 7'-11 1/8")

Fasteners

NET	ORDER	UNIT	ITEM	WHY
1	**1**	box	TimberTech CONCEALoc hidden fastener 500 ct (screws incl.)	168 needed · 500 supplied
1	**1**	box	Starborn Cap-Tor xd 2-3/4" Epoxy Driftwood 350 ct	106 screws needed (first row 7 · border 58 · fascia 41)

Fascia

NET	ORDER	UNIT	ITEM	WHY
3	**4**	ea	TimberTech Reserve Fascia Driftwood 0.58x11.95x12	+1 (left fascia 11'-6 1/16" · right fascia 11'-6 1/16" · front fascia 7'-10 3/16")

Rail

NET	ORDER	UNIT	ITEM	WHY
4	**4**	ea	TimberTech Fulton Rail 6' x 36" level panel Black	exact (cut to 5'-3 3/4")
1	**1**	ea	TimberTech Fulton Rail 8' x 36" level panel Black (pre-assembled steel balusters, cut to bay)	exact (cut to 7'-1 1/2")
2	**2**	ea	TimberTech Fulton 2" END post 36" Black w/ brackets, cap, skirt	exact
2	**2**	ea	TimberTech Fulton 2" LINE post 36" Black w/ brackets, cap, skirt	exact
1	**1**	ea	Rust-Oleum flat black touch-up (cut ends)	exact

Shade sail

NET	ORDER	UNIT	ITEM	WHY
1	**1**	ea	Shade sail — 4-corner shade sail, ~73 SF, Sand — SUPPLIED BY THE OWNER (GSX hangs and tensions it)	owner-supplied — not ordered
1	**1**	lot	Sail posts (2) — 4" steel, Black, on base plates on top of the deck, the rail corner posts at the front corners	owner's price
1	**1**	ea	2x10x8 #1 True Frame — solid blocking under each sail post base plate (rim to PF joist, doubled)	4 blocks
8	**10**	ea	1/2" x 8" HDG through-bolt + nut + washers — base plates through the decking into the blocking	+2 (4 per post)

Site

NET	ORDER	UNIT	ITEM	WHY
64	**64**	SF	Demo existing deck + haul-off	labor line — field verify the existing deck

FASTENER & CONNECTOR SCHEDULE

Knee braces	4x4 DF #1 at 45°, 2'-6" legs, both directions at every post: one along the beam toward mid-span, one toward the house up to a 2x10 block between the joists; (1) 1/2" through-bolt each end, 2" min. edge distance
Ledger	2x10 ledger x 7'-9" total · LedgerLOK 2 rows 2" from top/bottom, staggered 12" OC, first 3" from each end · 9 fasteners
Beam	(2)4x plies through-bolted with 1/2" HDG bolts, 2 rows 2" from top and bottom, staggered @ 24" OC, a bolt pair 6" from each end; CCQ cap with SDS screws at each post

Joist tape	G-Tape 2" every joist, PF joist, blocking and the ledger top · 4" on both rims and the beam(s)
Field	CONCEALoc at every joist in every gap (168)
Face screws	Cap-Tor xd color-matched ONLY at: first row 1 per joist on the house edge · borders 2 per bearing point @ 16" · fascia 2 at each board end then alternate top/bottom every 12", pre-drilled
Rail	Fulton 36" — 6 posts (2 CORNER, 2 END, 2 LINE); the 2 front corners are the shade-sail posts, base-plated on the deck — the panels bracket to them inside the outer rim ply, inner ply pocketed 2" wide, (2) 7/16" x 4-1/2" bolts per post

▼ Cut list

MEMBER	STOCK	LENGTH	QTY	NOTE
Field joists	2x10	10'-9"	5	@ 16" OC, crown up
Picture-frame joists	2x10	10'-9"	2	centre 5 1/4" from each frame face
Side rim plies	2x10	10'-9"	4	2-ply, laminated on the ground with RSS 2 rows @ 12" staggered
Ledger	2x10	7'-9"	1	top at deck height less board thickness; membrane behind, flashing over
Blocking	2x10	1'-2 1/2"	6	1 row at 5'-6" from the wall
Field boards run A	Reserve Driftwood	7'-0"	24	boards run parallel to the house
FRONT FLUSH (2)4x10 DF#1 beam	4x10	7'-9"	2	y 11'-2"; posts at 9" / 7'-3"
Posts under FRONT FLUSH (2)4x10 DF#1	6x6	8'-9 5/16"	2	field-measure each
Knee braces	4x4	3'-9 15/16"	4	45° both ends, 2'-6" legs · along the beam toward mid-span + up to a block between the joists
Left border	Reserve Driftwood square	11'-6 9/16"	1	mitre at the corners; dividers on the doubled joist
Right border	Reserve Driftwood square	11'-6 9/16"	1	mitre at the corners; dividers on the doubled joist
Front border	Reserve Driftwood square	7'-11 1/8"	1	mitre at the corners; dividers on the doubled joist
Left fascia	fascia	11'-6 1/16"	1	flush under the board nose; full board at the front corners, short piece at the house
Right fascia	fascia	11'-6 1/16"	1	flush under the board nose; full board at the front corners, short piece at the house
Front fascia	fascia	7'-10 3/16"	1	flush under the board nose; full board at the front corners, short piece at the house

THE STACK — COST + MARGIN sold at $13,000 · internal

Materials (raw)	$4,824.65

Materials × 1.15	$5,548.35
Sales tax 4.25% on factored materials	$235.80
Labor (rate card)	$2,704.00
Work — carries the margin	$8,488.15
General conditions at cost	$368.00
Cost	$8,856.15
SOLD — check or ACH	**$13,000** · $141.30/SF
Sell at the 40% floor	$15,342
Financed	**$13,978** · $162/mo at 6.99% × 10 yr
Gross profit	$4,144
Margin on the sold price (gross profit ÷ sell, GC at cost)	**31.9%**
Margin on the work (the GM the engine prices with)	32.8%
Markup on cost	46.8%
Commission 10% of sell	$1,300
Gross profit after commission	**$2,844** · 21.9% of sell

Labor

LINE	QTY	UNIT	RATE	EXT
Demolition — existing deck	64	SF	$2.00	$128.00
Diamond Pier install (replaces caisson)	2	pier	$50.00	$100.00
Frame	92	SF	$10.00	$920.00
Decking & fascia	92	SF	$8.00	$736.00
Railing	31	LF	$20.00	$620.00
Shade sail — 2 steel posts base-plated on the deck as the rail corners, blocking, house anchors, hang and tension	1	lot	$200.00	$200.00

General conditions — at cost, no margin

ITEM	AMOUNT	WHY
General conditions	$368	$4/SF x 92 SF

What the price includes — sell values

ITEM	VALUE
Demolition & haul-off	$176
Foundation & structure	$4,630
Decking & fascia	$4,026

ITEM	VALUE
Railing	$2,443
Shade sail posts + install — 2 steel posts at the rail corners (owner supplies the sail)	$1,384
Site & project services	$341
Total, check or ACH	**$13,000**
Total, financed	**$13,978**

THE QUOTE homeowner-facing · facts only

QUOTE · PRE-ENGINEERING

5549 S LANSING CT — ELEVATED DECK, 8' X 12' AT 10' (TIMBERTECH RESERVE DRIFTWOOD, FULTON BLACK, SHADE SAIL)

2x10 SYP frame at 16", (2)4x10 beam, 6x6 posts — on 2 Diamond Pier footings. TimberTech Reserve Driftwood decking, Fulton rail.

92 SF · 1 zone · **10'-0"** above grade, engineered · **30 psf** snow design

Prepared for Rachel Chiu · 5549 S Lansing Ct, Englewood

PRICE

Total investment $13,978 — financed, no money down

- 12 months no payments, no down payment
- 18 months no payments, no down payment
- 6.99% for 10 years — about $162/month

Check or ACH: $13,000 (7% savings)

WHAT THE PRICE INCLUDES

ITEM	
Demolition & haul-off	included
Foundation & structure	included
Decking & fascia	included
Railing	included
Shade sail posts + install — 2 steel posts at the rail corners (owner supplies the sail)	included
Site & project services	included
Total, check or ACH	**$13,000**
Total, financed	**$13,978**

Engineering. Engineer letter included.

INCLUDED — WHAT IT IS

ELEMENT	SELECTION
Decking	TimberTech Composite Reserve, Driftwood. Grooved boards, 3/16" gaps, matching fascia. Class C, not WUI; 30-yr product / 30-yr fade & stain.
Railing	TimberTech Fulton Rail, Black, 36". 31 LF.
Frame	7 2x10 #1 True Frame Joist SYP GC @ 16" OC x 10'-9"; 2-ply 2x10 sides; front rim = flush beam; FRONT FLUSH (2)4x10 DF#1: 7'-9", 2 posts at 9" / 7'-3" (worst 1,794 lb).
Foundation	2 Diamond Pier foundations (DP-50/50) — no excavation.
Design load	30 psf snow, stamped by a Colorado engineer.

ELEMENT	SELECTION
Hardware	Simpson ZMAX galvanized connectors. CONCEALoc fasteners.
Site	The existing deck removed and hauled off.

PLAN

7'-9" x 11'-5 1/2" outside to outside · 92 SF · 10'-0" above grade · 31 LF rail

SCHEDULE

1. Engineering & ordering (2–3 weeks). Stamped drawings, materials ordered, samples approved.
2. Day 1 — protection, utilities located (811), old deck out.
3. Day 2 — layout, 2 Diamond Piers driven, ledger set.
4. Days 3–4 — posts and beams set.
5. Days 5–6 — frame, flashing at every wall.
6. Days 7–8 — decking, borders first.
7. Days 9–9 — Fulton rail.
8. Day 10 — final clean; end of construction.

NEXT STEP

Next step: we will get the engineer letter.

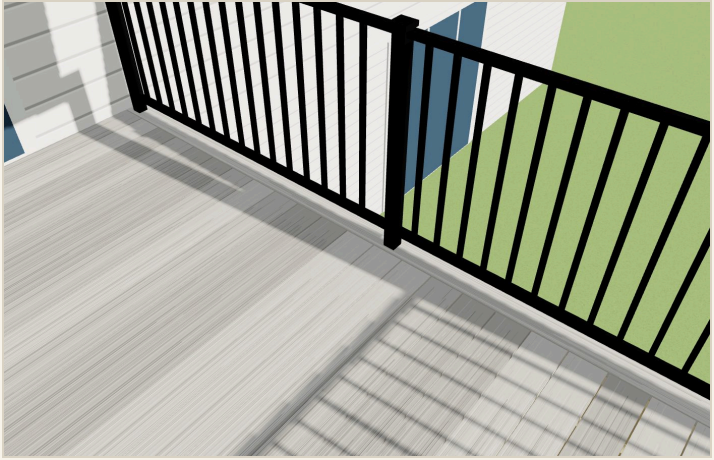
RENDERS 3D model · design intent · frame shown as specified



From the yard



From the front corner



Standing on the deck



Under the deck — posts, beams, footings

SUN AND SHADE the real sun at this address · the covers and the blinds through the day



Jul 15 · 9 AM — sun E, 35° up



Jul 15 · 12 PM — sun SE, 67° up



Jul 15 · 3 PM — sun WSW, 60° up

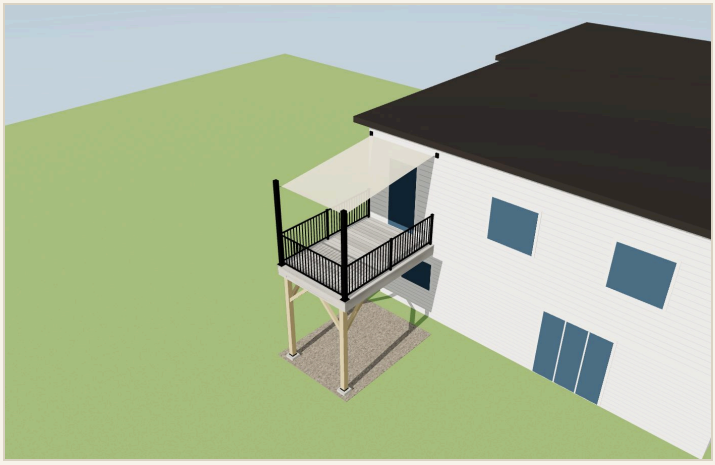


Jul 15 · 6 PM — sun W, 26° up

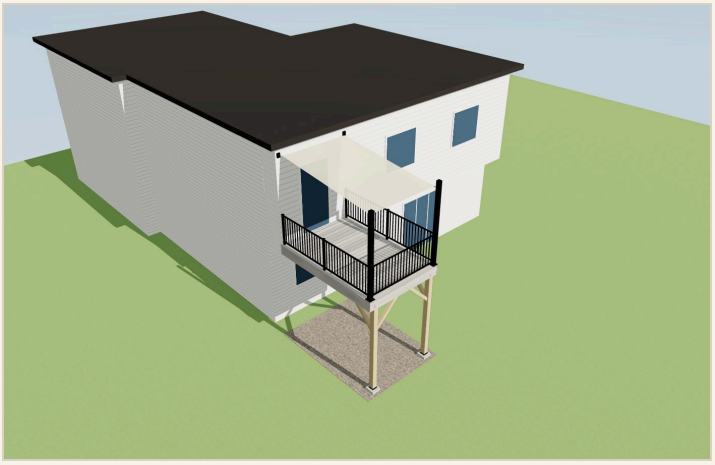


Dec 21 · 3 PM — sun SW, 14° up

ISOMETRICS



Isometric from the stair corner



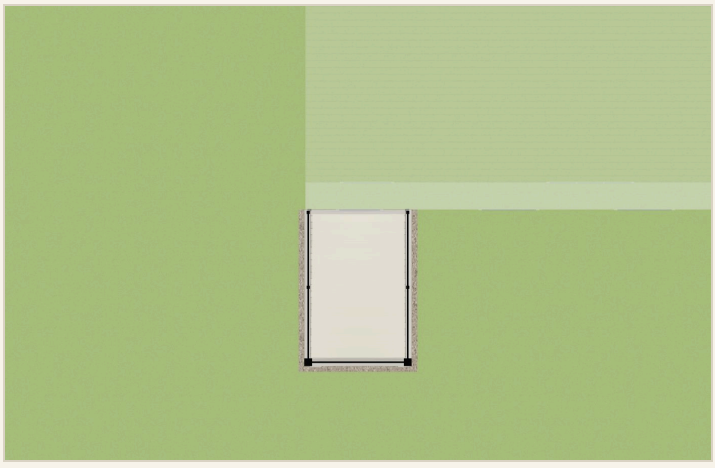
Isometric from the open corner



Low isometric

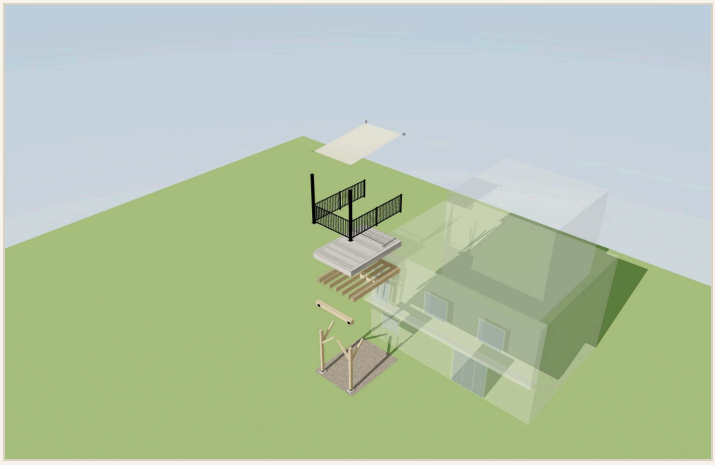


Isometric from under the frame

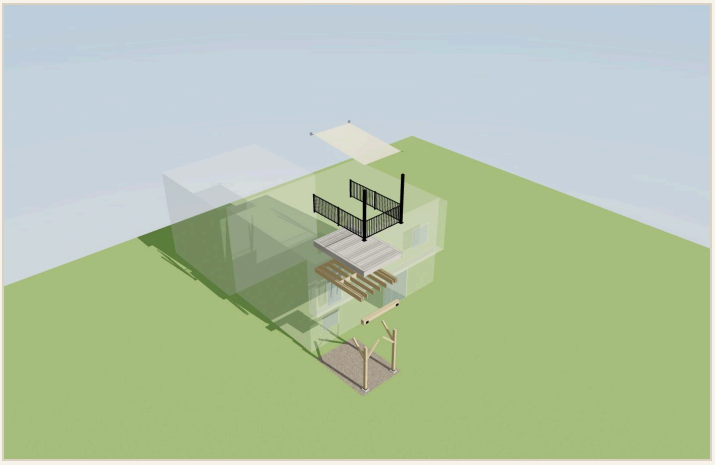


Plan from above

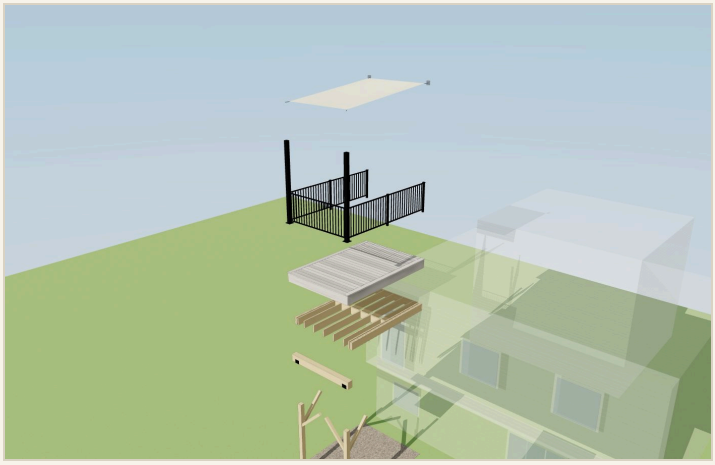
EXPLODED



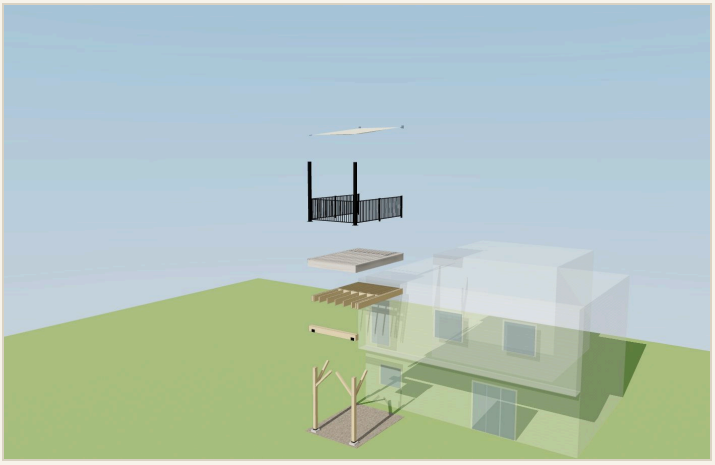
Exploded — every layer



Exploded — from the open corner



Exploded — from the front corner



Exploded — low angle

G-001 General notes · design criteria

G-001 GENERAL NOTES · DESIGN CRITERIA

Loads, codes, structure summary, sheet index, site — the permit-set cover

5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

GSX · Jade Pender 303-550-9558 · October 19, 2026 · Rev 0 FOR REVIEW

PROJECT

5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech Reserve Driftwood, Fulton black, shade sail)
5549 S Lansing Ct, Englewood CO 80111
Prepared for Rachel Chiu
92 SF · 10'-0" above grade · 1 zone

DESIGN CRITERIA

Deck live load	40 psf uniform (IRC Table R301.5); guards 200 lb concentrated / 50 plf
Ground snow	30 psf (ground; deck design uses the larger of snow and live) — 40 psf live governs; 30 psf ground snow, roof 30 psf non-reducible (Arapahoe County design criteria); deck tables at 40 psf live + 10 psf dead
Dead load	10 psf deck
Roof snow (cover)	—
Design total	50 psf on the deck
Wind	115 mph Vult (3-second gust), Exposure B (suburban), Risk Cat. II — county design-criteria table (2015 cycle; confirm the 2021 table)
Frost depth	3'-0" (footings bear below)
Soil bearing	1,500 psf presumptive (IRC R401.4.1) — no soils report
Seismic	B
Wildfire (WUI)	not in a WUI zone
Codes	2021 IRC / IBC as adopted by Arapahoe County (Res. 21-394, eff. 4/1/2022); IRC R507 decks with the county-amended Colorado ICC Deck Building Guide; ASCE 7-16 loads; Tables R507.5 / R507.6 / R507.4 / R507.9
Engineering	Stamped structural set by a Colorado PE required — the stamped set governs

STRUCTURE

Footings	DP-50/50
Posts	2 x 6x6 Douglas fir #1 with 4x4 knee braces
Beams	FRONT FLUSH (2)4x10 DF#1: 7'-9", 2 posts at 9" / 7'-3" (worst 1,794 lb)
Joists	7 2x10 #1 True Frame Joist SYP GC @ 16" OC x 10'-9"
Rims	2-ply 2x10 sides; front rim = flush beam
Ledger	2x10 ledger x 7'-9" total · LedgerLOK 2 rows 2" from top/bottom, staggered 12" OC, first 3" from each end · 9 fasteners
Decking	TimberTech Reserve Sand — 24 rows parallel to the house, picture frame in Reserve Driftwood, 3/16" gaps, CONCEALoc
Rail	Fulton 36" Black: 5 bays, 6 posts, 31 LF

SHEET INDEX

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D-501	Details
T-400	Build steps (build set)
T-700	Order + schedules (takeoff)

SITE

Rear / front lot line: deck edge 13.0' from the line, setback per the PUD...
Side lot line: deck edge 23.7' from the line, 5' required
Easement: Check plat R1535678 for rear / side utility and drainage...
Unincorporated Arapahoe County (Cherry Creek CDP; Englewood mailing address). Parcel 2075-14-4-05-001 / PIN 031772427, Lot 41 Blk 9 Cherry Creek Vista 6th Flg Amend Resub (plat R1535678), 6,401 SF corner lot (front S Lansing Ct east, street side E Berry Ave north). R-P PUD Cherry Creek Vista (Z75-030 / P75-027 / P75-028). Deck on the WEST (rear) wall, north end, hung on the 2' second-floor cantilever that runs the full 33' wall (assessor sketch). Tax: CO 2.9 + RTD 1.0 + CD 0.1 + Arapahoe 0.25 = 4.25% (no city).

FLAGS (INTERNAL — remove before the permit submittal)

ENGINEER	FRONT FLUSH (2)4x10 DF#1: (2)4x10 not in IRC R507.5 — treated as...
ENGINEER	Shade sail: the sail posts stand on the deck as the rail corners...
ENGINEER	Shade sail: sail wind load (115 mph) on the 2 steel posts and the...
ENGINEER	house floor cantilevers past the foundation — ledger attachment...
ENGINEER	Arapahoe County deck guide: a ledger 'shall not be attached to a...
CODE	county guide: pier footings 12" x 36" minimum unless engineered —...
CODE	guards above 30": 36" min., openings under 4", 200 lb point load...
CODE	county submittal: site plan with lot / block and distances to...

No scale

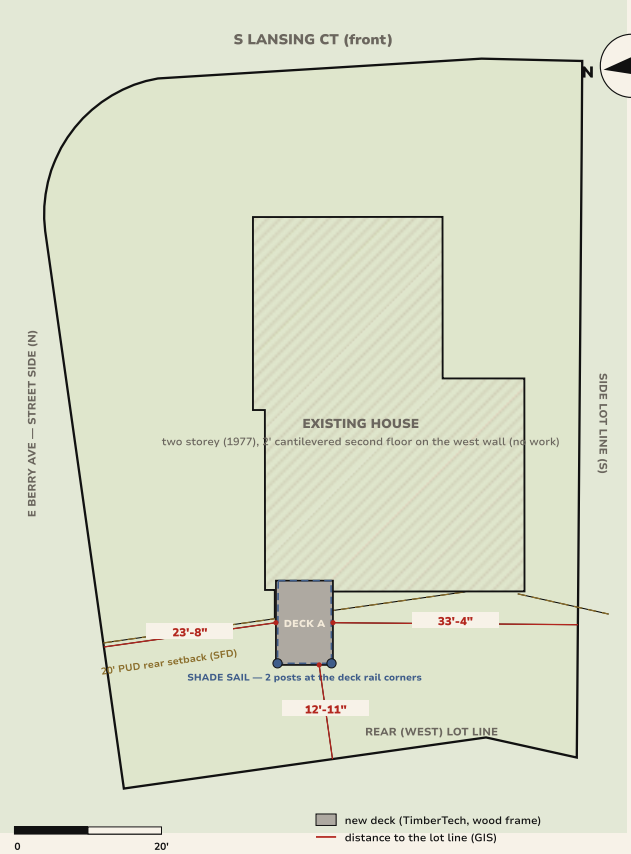
Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

C-100 SITE PLAN

Parcel, existing house, the new deck, distances to the lot lines, setbacks

5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

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PARCEL

5549 S Lansing Ct, Englewood CO 80111
Parcel 2075-14-4-05-001 · PIN 031772427
Lot 41, Block 9, Cherry Creek Vista 6th Filing Amended Resub (plat R1535678)
Unincorporated Arapahoe County · zoning R-P PUD (Z75-030 / P75-027 / P75-028)
Lot 6,401 SF (GIS) · corner lot · flood zone X
House 1977, 2 storey, 1,704 SF + 660 SF basement + 441 SF garage (assessor)
HOA: Cherry Creek Vista II

LOT COVERAGE

Lot area	6,401 SF
Lot coverage limit	none in the PUD plan
Existing deck (assessor)	48 SF (8' x 6') — removed
New deck	8' x 12' · ~92 SF of frame, 10' above grade

SETBACKS

Elevated decks get no setback projection allowance (Arapahoe LDC) — the deck must meet the setbacks
GIS puts the new deck front ~13' off the west line: passes the 10' standard, not the 20' — confirm the governing column and the rear line (corner lot) with Zoning before submittal
E Berry Ave street side 10' required — ~23'-8" clear; south side 5' — ~33' clear
Measured (red): to the nearest lot line from the new work — GIS, not survey

SITE WORK

· Remove the existing 8' x 6' deck and its two posts / pads; haul off
· Two Diamond Pier DP-50 foundations (no excavation spoils), 6x6 DF #1 posts with 4x4 knee braces
· New ledger on the 2' cantilevered second floor per the engineer's detail; membrane + Z-flash, (4) DTT1Z lateral ties
· No stairs — the deck is reached from the house door only
· Call 811 before the piers; protect the lawn and the patio slab at the north wall

Arapahoe County GIS (address point, parcel, 2024 roof prints, R-P zoning, jurisdiction, flood, wildfire layers), county assessor record and sketch, 2nd Amended PDP P75-027 setback table, county deck permit guide (rev. 8/20/2024) and the county-amended ICC deck guide. GIS lines are not a survey — an ILC governs if Zoning asks.

Plan · approx. 1" = 20' at full size · GIS lines, not a survey

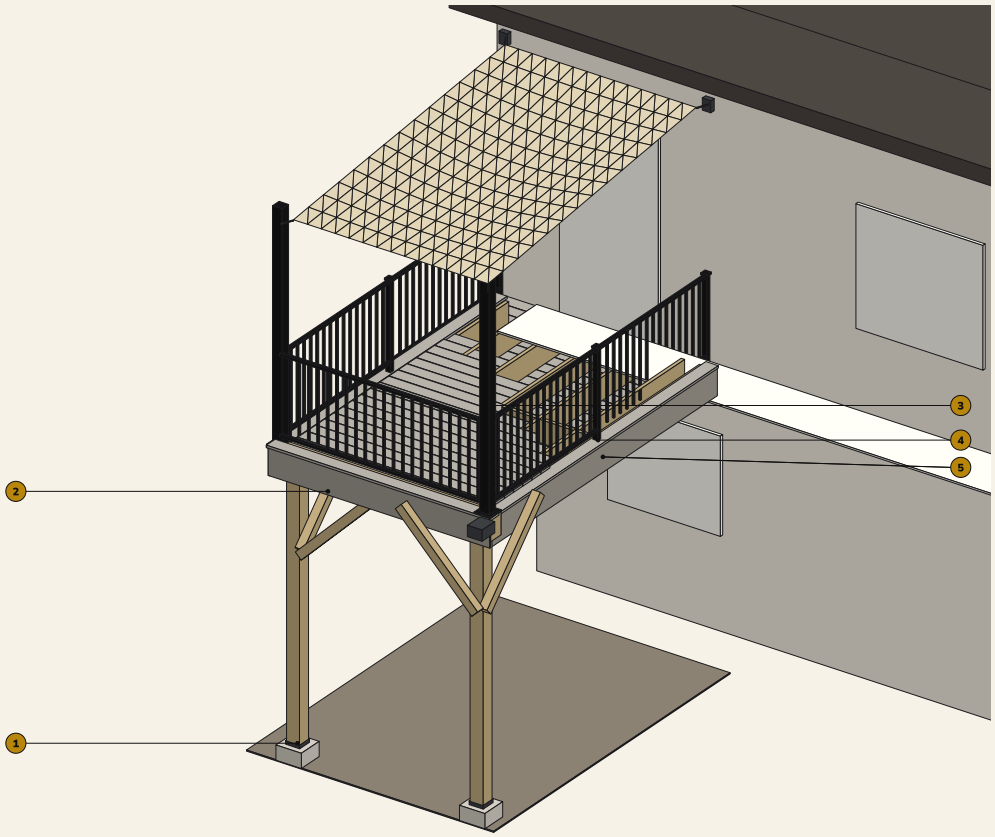
Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

A-002 ISOMETRIC — ASSEMBLED, LABELED

Everything you see called out: piers, decking, picture frame, skirt, stair, roof covers, finishes

5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

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KEY

1. Diamond Pier DP-50/50 — head + 4 driven pins (frost-protected, no excavation)
2. 4x4 DF #1 knee brace at 45°, both directions, through-bolted
3. TimberTech Reserve Driftwood field decking, hidden fasteners
4. Picture frame + breaker boards — TimberTech Reserve Driftwood
5. Fascia

House shown for context only (the deck work only). The frame under the decking is called out on S-102 and A-003.

Axonometric · not to scale · the model the plans are drawn from

Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

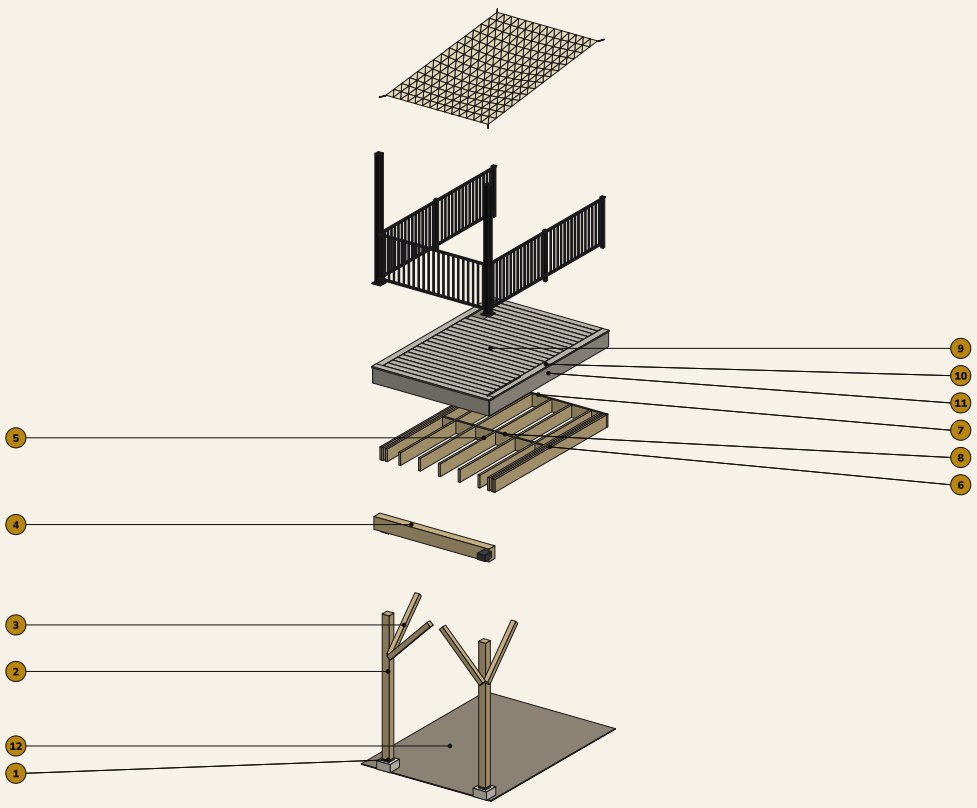
A-003 Isometric — exploded, labeled

A-003 ISOMETRIC — EXPLODED, LABELED

Each build layer lifted in order: piers · posts + knee braces · beams · frame · decking · rail

5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

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KEY

1. Diamond Pier DP-50/50 — head + 4 driven pins (frost-protected, no excavation)
2. 6x6 Douglas fir #1 post on a ABA66Z standoff base
3. 4x4 DF #1 knee brace at 45°, both directions, through-bolted
4. Beam
5. 2x10 joist
6. Rim
7. Ledger
8. Blocking
9. TimberTech Reserve Driftwood field decking, hidden fasteners
10. Picture frame + breaker boards — TimberTech Reserve Driftwood
11. Fascia
12. Gravel over 6-mil poly under the decks

Layers lifted in build order. Lifts are for clarity only; every member returns straight down to its place.

Axonometric · not to scale · the model the plans are drawn from

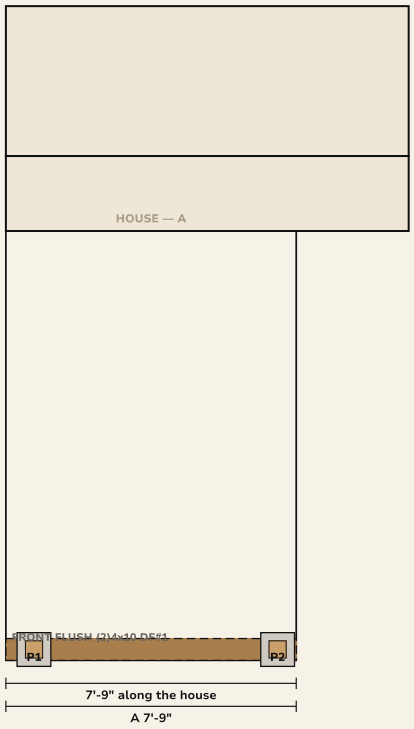
Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

S-100 FOUNDATION PLAN

Footings, posts and beam lines with the footing schedule; beams dashed (below the joists)

5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

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FOOTING SCHEDULE

Diamond Pier DP-50/50 (head + 4 x 50" pins), 3,300 lb allowable

ID	X	Y	ZONE	LOAD
P1	9"	11'-2"	A	1,794 lb
P2	7'-3"	11'-2"	A	1,794 lb

x from the left frame face facing the house; y from the reference wall.
Frost 3'-0" · soil 1,500 psf · worst post 1,794 lb

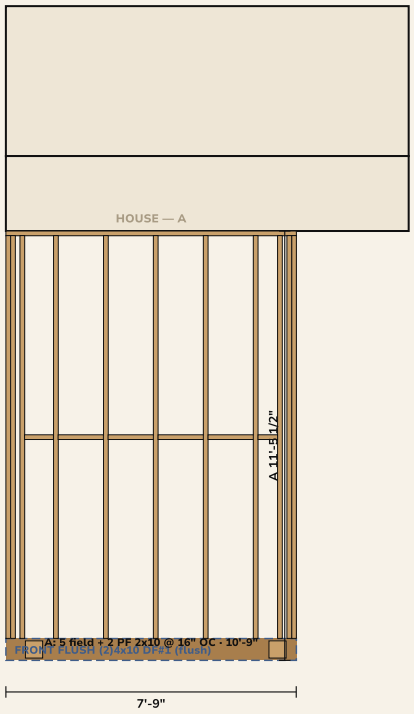
Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

S-101 FRAMING PLAN

Ledger, joists, rims, blocking, beams; every member in the model — the cut list is on T-600

5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

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FRAMING NOTES

7 2x10 #1 True Frame Joist SYP GC @ 16" OC x 10'-9"

2-ply 2x10 sides; front rim = flush beam

2x10 ledger x 7'-9" total · LedgerLOK 2 rows 2" from top/bottom, staggered 12" OC, first 3" from each end · 9 fasteners

blocking one row over the beam, staggered

G-Tape 2" every joist, PF joist, blocking and the ledger top · 4" on both rims and the beam(s)

FRONT FLUSH (2)4x10 DF#1: 7'-9", 2 posts at 9" / 7'-3" (worst 1,794 lb)

Connectors:

1/2" x 10" HDG hex bolt + nut + 2 washers — knee braces (brace to the 6x6 post): 4

1/2" x 12" HDG hex bolt + nut + 2 washers — knee braces (brace to the (2)4x beam): 2

1/2" x 6" HDG hex bolt + nut + 2 washers — knee braces (brace to the joist block): 2

Simpson LUS28Z face-mount hanger 2x8/2x10: 14

Simpson HUCQ210-2-SDS concealed-flange double 2x10 hanger (SDS screws included): 4

Simpson DTT1Z lateral tie kit w/ SDS screws: 4

Simpson CCQ column cap, 7" seat for the (2)4x10 beam on a 6x6 post (SDS screws incl.) — confirm SKU / engineer's equal:

1/2" x 8" HDG carriage bolt + nut + washer: 10

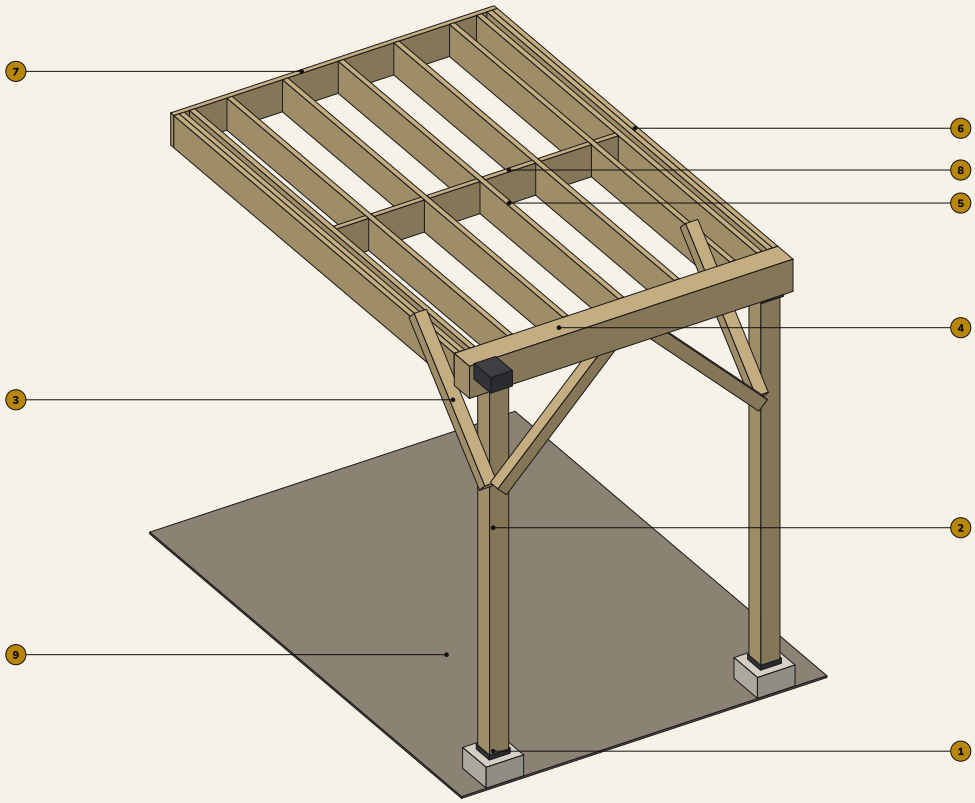
Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

S-102 STRUCTURE ISOMETRIC — FRAME, LABELED

Piers, post bases, posts, knee braces, post caps, beam, joists, double rims, ledger, blocking

5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

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KEY

1. Diamond Pier DP-50/50 — head + 4 driven pins (frost-protected, no excavation)
2. 6x6 Douglas fir #1 post on a ABA66Z standoff base
3. 4x4 DF #1 knee brace at 45°, both directions, through-bolted
4. Beam
5. 2x10 joist
6. Rim
7. Ledger
8. Blocking
9. Gravel over 6-mil poly under the decks

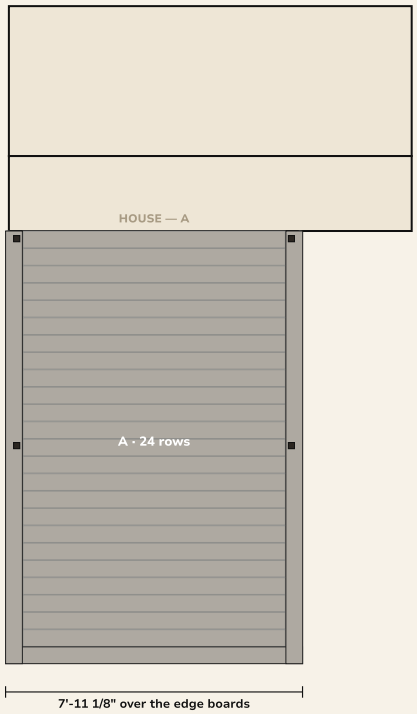
Decking, rail and house removed to show the frame. Joists hang from the ledger in face-mount hangers, bear on the drop beam with H2.5AZ ties; double rims in concealed-flange hangers.

A-101 DECKING PLAN

Every board, border and divider as it lays; rail posts shown for the pockets

5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

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DECKING

Field: TimberTech Reserve Driftwood, 5.36" x 0.94", 3/16" gaps

Border: Reserve Driftwood

Boards run parallel to the house; edge overhang 1 1/16" over the fascia

CONCEALoc at every joist in every gap (168)

Cap-Tor xd color-matched ONLY at: first row 1 per joist on the house edge · borders 2 per bearing point @ 16" · fascia 2 at each board end then

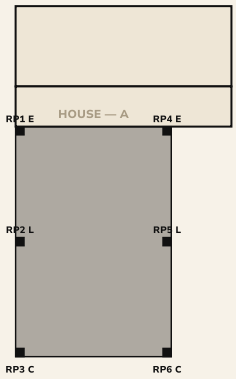
Rows per zone: A 24 (no rips)

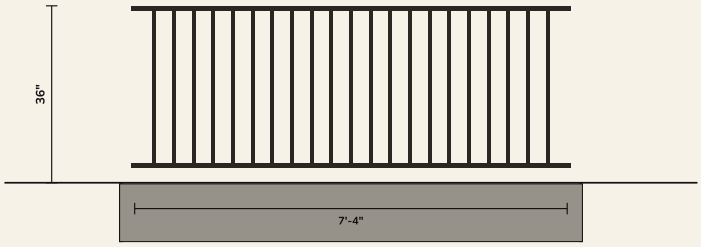
A-201 RAILING PLAN + ELEVATION

Post IDs and types (E end · L line · C corner), bay lengths, outer-edge elevation

5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

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RAILING

Fulton 36" Black: 5 bays, 6 posts, 31 LF

Fulton 36" — 6 posts (2 CORNER, 2 END, 2 LINE); the 2 front corners are the shade-sail posts, base-plated on the deck — the panels bracket to them inside the outer rim ply, inner ply pocketed 2" wide, (2) 7/16" x 4-1/2" bolts per post

Bays: left 5'-6 1/4" · left 5'-6 1/4" · right 5'-6 1/4" · right 5'-6 1/4" · front 7'-4"

Guard 36" min above the deck; openings under 4" (IRC R312). Rail LF for labor = deck-edge LF of actual rail.

Plan above · elevation of the outer edge below

Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

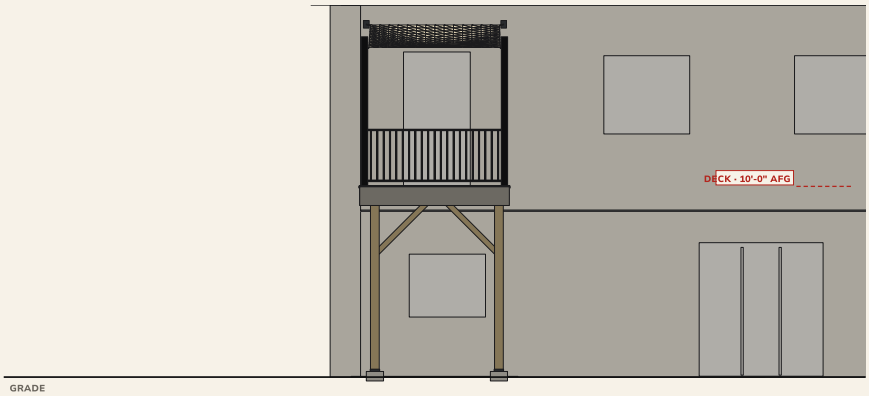
A-202 ELEVATIONS

Front from the yard and both ends, drawn from the model — deck and roof-cover heights

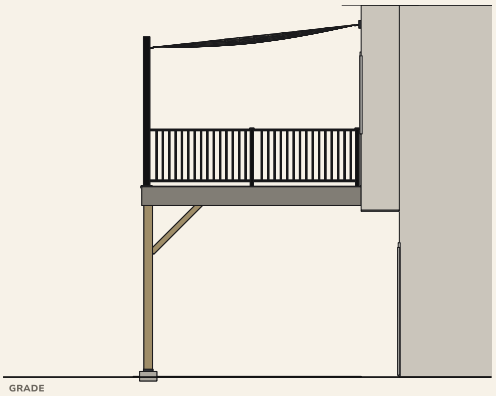
5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

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FRONT ELEVATION — from the yard



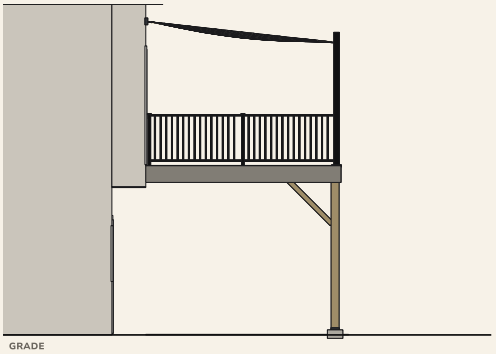
RIGHT END ELEVATION



NOTES

· Deck surface 10'-0" above grade
· Heights AFG = above finished grade at the beam line; field-verify grade before setting the piers
· Deck frame: 2x10 #1 True Frame GC @ 16" OC on a (2)4x10 DF #1 drop beam, 2 posts with 4x4 knee braces both ways
· Guard 36" min. on every open side — the surface is 10'-0" above grade (IRC R312.1)

LEFT END ELEVATION



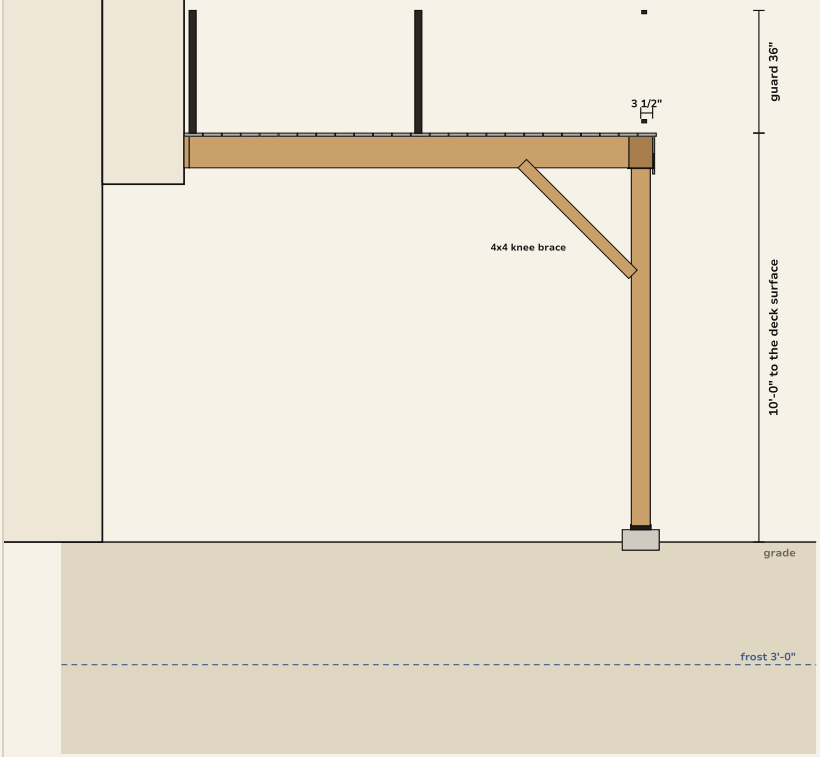
A-301 SECTION + FRONT ELEVATION

Cut through the deepest zone; heights to grade, frost line, cantilever, guard

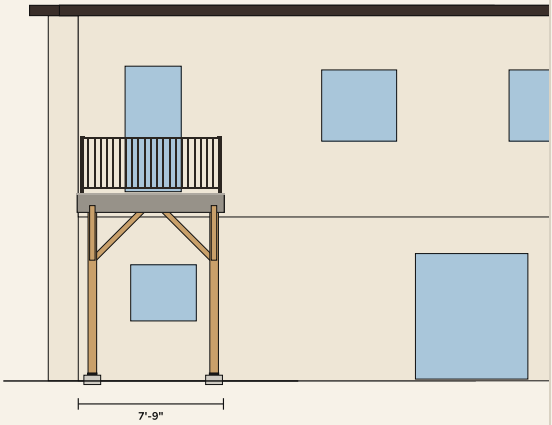
5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

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SECTION through zone A at 3'-10 1/2" — looking toward the left end



FRONT ELEVATION — from the yard



SECTION NOTES

Deck surface 10'-0" above grade; guard 36" where the drop exceeds 30"

FRONT FLUSH (2)4x10 DF#1: 7'-9", 2 posts at 9" / 7'-3" (worst 1,794 lb)

Posts 2 x 6x6 Douglas fir #1 with 4x4 knee braces on DP-50/50

2x10 ledger x 7'-9" total · LedgerLOK 2 rows 2" from top/bottom, staggered 12" OC, first 3" from each end · 9 fasteners

G-Tape 2" every joist, PF joist, blocking and the ledger top · 4" on both rims and the beam(s)

Section left · elevation right · not to a common scale

Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

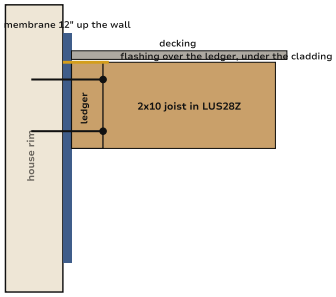
D-501 DETAILS

Ledger · post / beam / cap · footing · rail post · deck edge · the job's special condition

5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

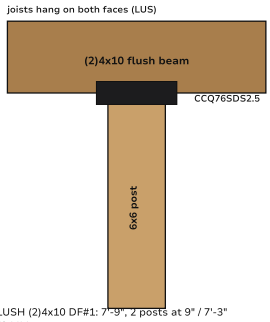
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1 LEDGER AT THE HOUSE



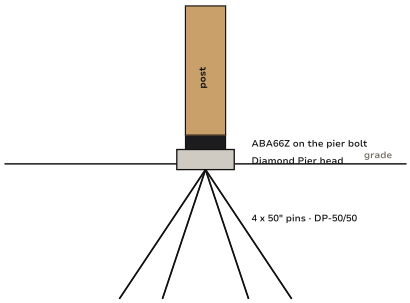
2x10 ledger x 7'-9" total · LedgerLOK 2 rows 2" from top/bottom, staggered 12" OC, first 3" from each end · 9 fasteners

2 POST · BEAM · CAP

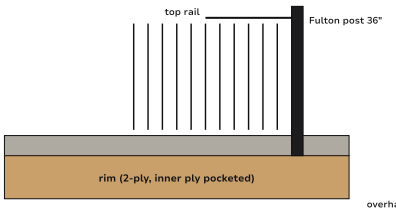


FRONT FLUSH (2)4x10 DF#1: 7'-9", 2 posts at 9" / 7'-3" (worst 1,794 lb)

3 FOOTING · POST BASE

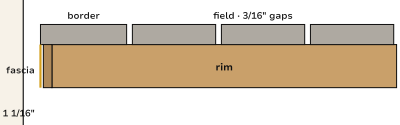


4 RAIL POST · GUARD



Fulton 36" — 6 posts (2 CORNER, 2 END, 2 LINE); the 2 front corners are the shade-sail posts, base-plated on the deck — the panels bracket to them inside the outer rim ply, inner ply pocketed 2" wide, (2) 7/16" x 4-1/2" bolts per post

5 DECK EDGE · BORDER



CONCEALoc at every joist in every gap (168)

6 FLUSH BEAM

Flush beam: top of beam = top of joists. Joists hang on both faces in LUS hangers; opposing hangers per Simpson and the stamped set.

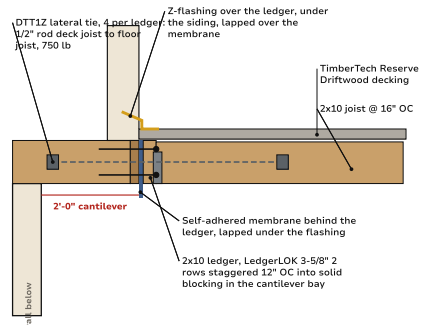
D-502 CONNECTION DETAILS — FRAME

Ledger on the cantilever · hangers · joists on the beam · (2)4x10 on the cap · pier + base · knee braces

5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

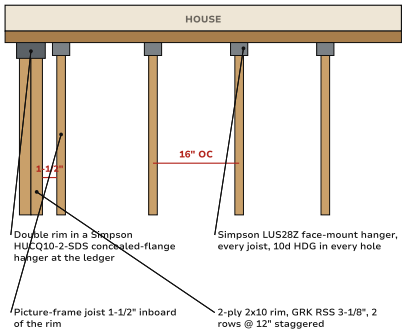
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1 LEDGER ON THE CANTILEVERED FLOOR — SECTION

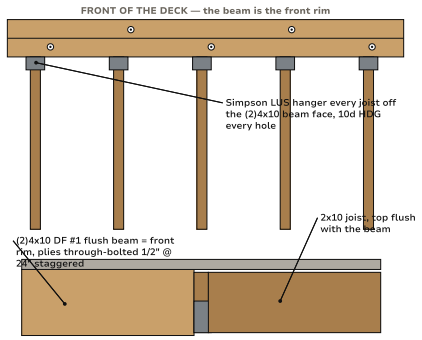


ENGINEER: Arapahoe County bars a ledger on a cantilevered floor without engineer approval. The stamped detail governs: ledger into the blocked cantilever joists, or a freestanding beam on posts at the house.

2 HANGERS — PLAN AT THE LEDGER + SIDE RIM

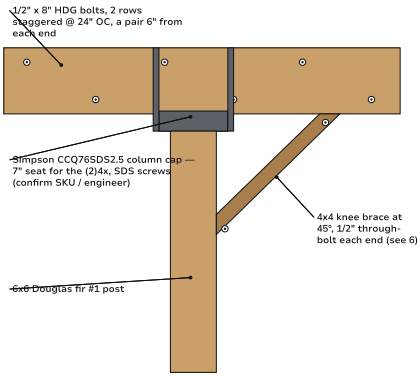


3 JOISTS INTO THE FLUSH BEAM — PLAN + SECTION

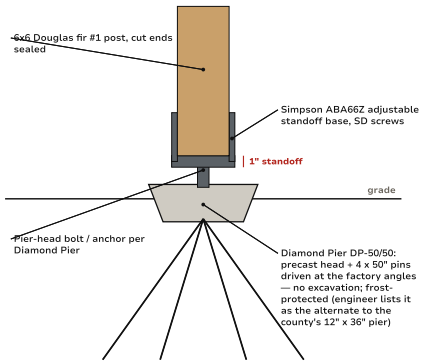


No cantilever: the beam carries the front edge, joists hang both ends

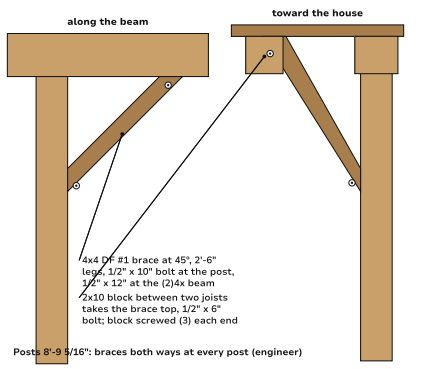
4 (2)4x10 BEAM ON THE 6x6 — ELEVATION



5 POST BASE ON THE DIAMOND PIER — SECTION



6 KNEE BRACES — BOTH DIRECTIONS



Posts 8'-9 5/16": braces both ways at every post (engineer)

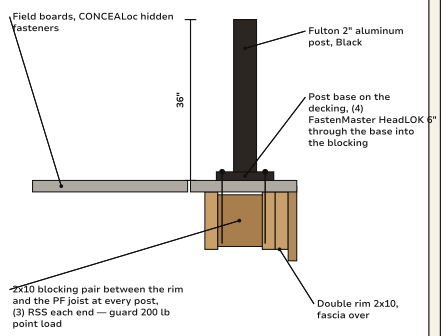
D-504 CONNECTION DETAILS — DECK EDGE, RAIL, LATERAL

Rail post mounting · picture frame + fascia · guard bay · DTT1Z lateral ties · joist tape · fastening schedule

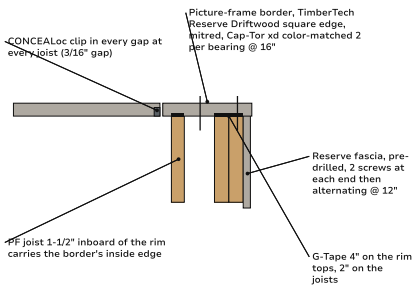
5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

GSX · Jade Pender 303-550-9558 · October 19, 2026 · Rev 0 FOR REVIEW

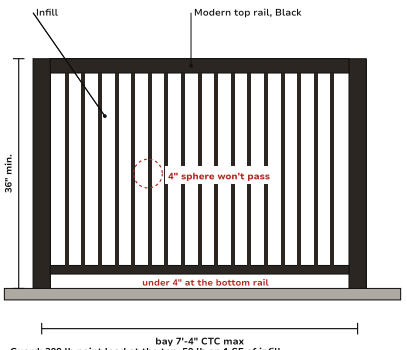
1 RAIL POST AT THE RIM



2 DECK EDGE — PICTURE FRAME + FASCIA

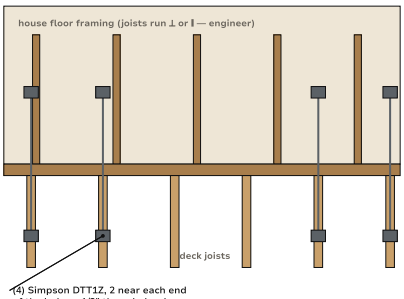


3 GUARD — ONE BAY, ELEVATION



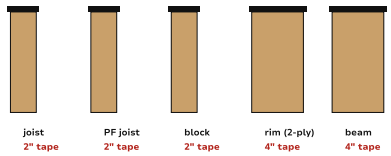
Guard: 200 lb point load at the top, 50 lb on 1 SF of infill (IRC Table R301.5) — submit the rail code report

4 LATERAL LOAD — DTT1Z TIES (PLAN)



(4) Simpson DTT1Z, 2 near each end of the ledger: 1/2" threaded rod through the ledger and the band into a floor joist (or blocking) inside — 750 lb each (IRC R507.9.2)

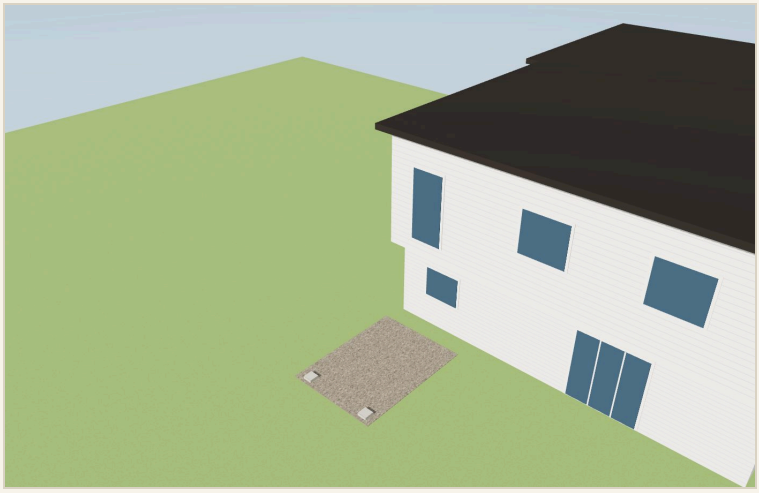
5 JOIST TAPE + BLOCKING



Nichigo G-Tape 3035 (black) on every top edge before decking, lapped over the ends and down the sides 1/2"; one row of 2x10 blocking over the beam, (3) structural screws each

6 FASTENING SCHEDULE

Ledger to house	LedgerLOK 3-5/8", 2 rows staggered 12" OC — engineer
Lateral	(4) DTT1Z, 1/2" rod into the floor framing
Joist to ledger	LUS28Z, 10d HDG every hole
Rim to ledger	HUCQ10-2-SDS, SDS screws
Rim plies	GRK RSS 3-1/8", 2 rows @ 12" staggered
Joist to beam	H2.5AZ each joist + rim, 10d x 1-1/2"
Beam plies	1/2" x 8" HDG bolts, 2 rows staggered @ 24"
Beam to post	CCQ76SDS2.5 cap, SDS screws
Post to pier	ABA66Z on the Diamond Pier, SD screws
Knee braces	1/2" through-bolt each end (x10 / x12 / x6)
Field boards	CONCEALoc, every joist, every gap
Borders / fascia	Cap-Tor xd color-matched, pre-drilled
Rail posts	(2) 7/16" bolts per post



Step 1 · Footings — 2 Diamond Pier DP-50/50 driven-pin footings

T-400 · Step 1 · Footings

GOES IN

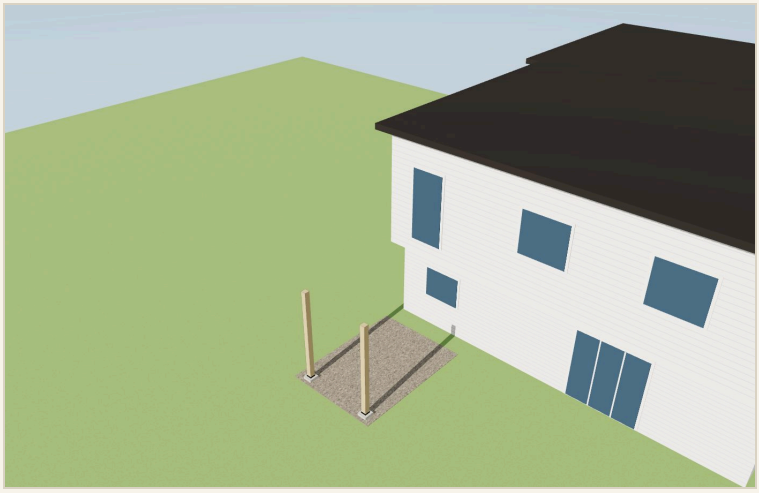
- (2) Diamond Pier DP-50/50 (head + 4 x 50" pins, plugs, caps)
- (2) Simpson ABA66Z adjustable post base 6x6 on the pier bolt

HOW

- Call 811 and locate before any hole or pin. Strip the existing deck first.
- Pull the beam lines from the house wall: FRONT FLUSH (2)4x10 DF#1 at 11'-2" from the reference wall, posts at 9" / 7'-3".
- Set 2 Diamond Pier heads level at grade on undisturbed soil and drive the 4 pins in each; caps on.

CHECK

- Footings on the layout within 1/2"; bearing on undisturbed soil
- Pins fully driven, heads level
- Photos before backfill



Step 2 · Posts — 6x6 Douglas fir #1 on ZMAX bases

T-401 · Step 2 · Posts

GOES IN

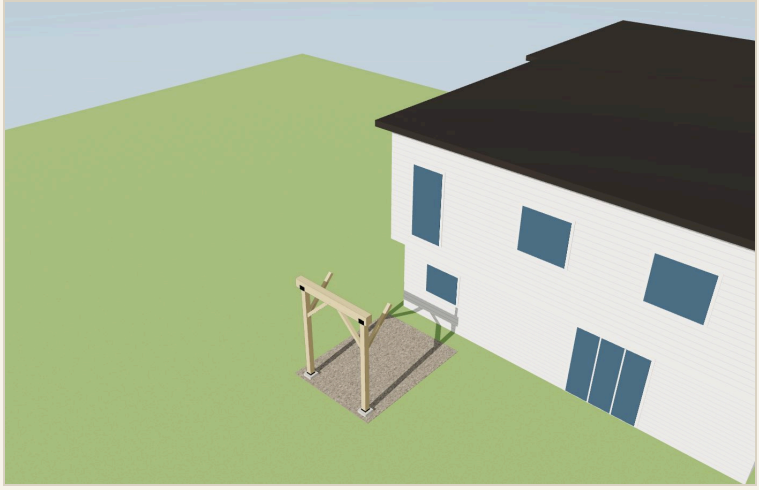
- (2) 6x6x10 Douglas Fir #1
- (2) Simpson ABA66Z adjustable post base 6x6 on the pier bolt
- (1) 2x10x8 #1 True Frame — solid blocking under each sail post base plate (rim to PF joist, doubled)

HOW

- 6x6 posts on the bases; cut so the beam top lands at the joist bottom (drop beam) or the joist top (flush beam): FRONT FLUSH (2)4x10 DF#1 posts 8'-9 5/16" — field-measure each.
- Plumb both ways, brace, then the caps go on with the beams.

CHECK

- Posts plumb to a string; tops on the beam line
- No post in ground contact without a standoff



Step 4 · Beams — FRONT FLUSH (2)4x10 DF#1 (2 posts)

T-403 · Step 4 · Beams

GOES IN

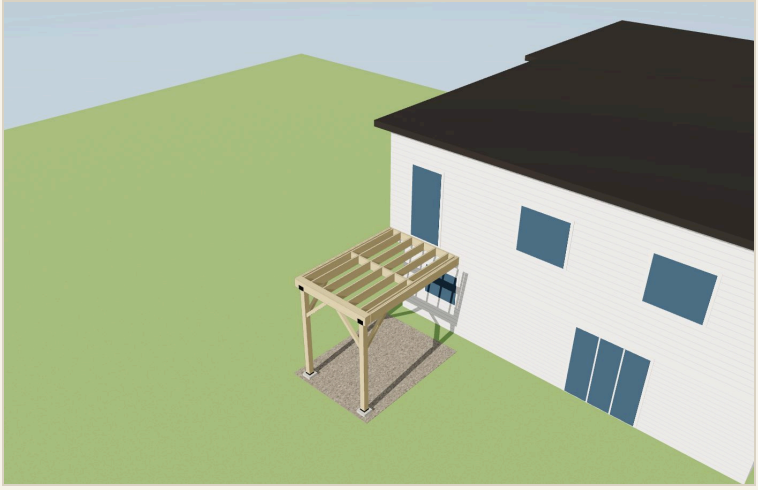
- (2) 4x10x8 Douglas Fir #1
- (2) 6x6x10 Douglas Fir #1
- (2) Simpson CCQ column cap, 7" seat for the (2)4x10 beam on a 6x6 post (SDS screws incl.) — confirm SKU / engineer's equal
- (2) 4x4x8 Douglas Fir #1 — knee braces
- (4) 1/2" x 10" HDG hex bolt + nut + 2 washers — knee braces (brace to the 6x6 post)
- (2) 1/2" x 12" HDG hex bolt + nut + 2 washers — knee braces (brace to the (2)4x beam)
- (2) 1/2" x 6" HDG hex bolt + nut + 2 washers — knee braces (brace to the joist block)
- (10) 1/2" x 8" HDG carriage bolt + nut + washer
- (8) 1/2" x 8" HDG through-bolt + nut + washers — base plates through the decking into the blocking

HOW

- FRONT FLUSH (2)4x10 DF#1: 7'-9" one piece, 2 posts @ 6'-6" max
- Crown up, strung straight, caps screwed off with the SDS screws in the box.
- Two-ply beam: through-bolt the plies with 1/2" HDG bolts, 2 rows 2" from top and bottom, staggered @ 24" OC, a pair 6" from each end — before it goes up.
- Knee braces once the beam is capped: 4x4 at 45°, 2'-6" legs, one along the beam toward mid-span and one toward the house up to a 2x10 block between two joists (set the block with the frame) — 1/2" through-bolt each end, then pull the temporary bracing.

CHECK

- Beam top = joist bottom (drop) / joist top (flush)
- Splices land on a post; caps fully fastened
- Every knee brace bolted both ends; posts still plumb



Step 5 · Frame — 2x10 joists at 16", rims, ledger, blocking over the beam

T-404 · Step 5 · Frame

GOES IN

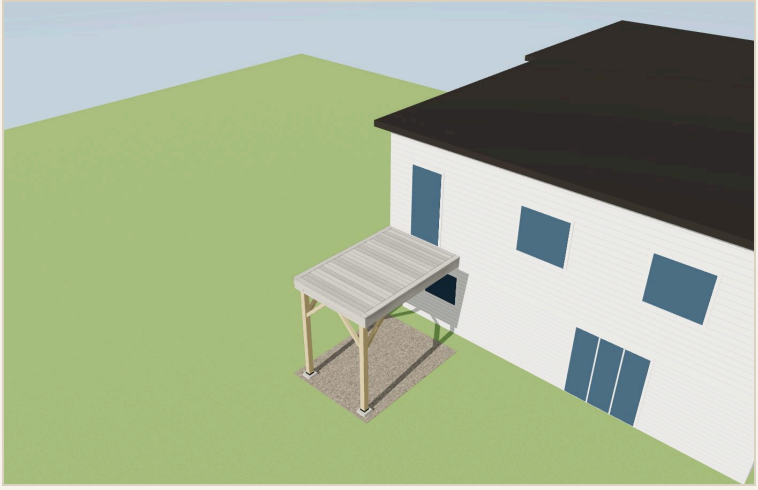
- (13) 2x10x12 #1 True Frame Joist SYP GC
- (14) Simpson LUS28Z face-mount hanger 2x8/2x10
- (4) Simpson HUCQ210-2-SDS concealed-flange double 2x10 hanger (SDS screws included)
- (1) Camo 9ga HDG hanger nail 3" 5 lb (~350)
- (1) Camo 9ga HDG hanger nail 1-1/2" 5 lb (~650)
- (1) FastenMaster LedgerLOK 3-5/8" 50 ct
- (1) Ledger flashing set: SA membrane 6"x75' + (2) 10' alum. Z + end dams
- (2) Nichigo G-Tape 3035 2" x 65' black
- (1) Nichigo G-Tape 3035 4" x 65' black

HOW

- 2x10 ledger x 7'-9" total · LedgerLOK 2 rows 2" from top/bottom, staggered 12" OC, first 3" from each end · 9 fasteners
- A: 5 joists @ 16" OC x 10'-9". Crown up, hangers at the ledger, both faces of the flush beam.
- Blocking one row over the beam, staggered.
- G-Tape 2" every joist, PF joist, blocking and the ledger top · 4" on both rims and the beam(s)

CHECK

- Frame square: diagonals within 1/8"
- Every hanger fully nailed; ledger fasteners per the schedule
- Tape continuous; photos of the ledger, hangers and footings before decking



Step 6 · Decking — TimberTech Reserve Driftwood, hidden fasteners

T-405 · Step 6 · Decking

GOES IN

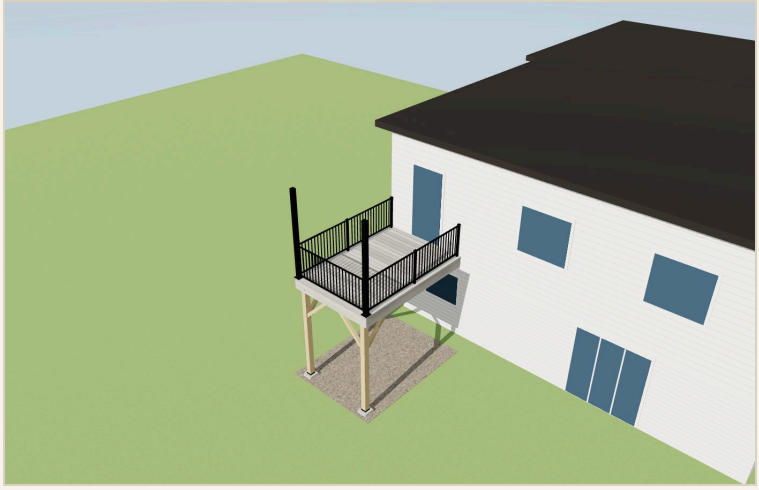
- (12) TimberTech Reserve Driftwood 1x6x16 Grooved
- (1) TimberTech Reserve Driftwood 1x6x12 Square Edge — borders
- (1) TimberTech Reserve Driftwood 1x6x20 Square Edge — borders
- (3) TimberTech Reserve Fascia Driftwood 0.58x11.95x12
- (1) TimberTech CONCEALoc hidden fastener 500 ct (screws incl.)
- (1) Starborn Cap-Tor xd 2-3/4" Epoxy Driftwood 350 ct

HOW

- CONCEALoc at every joist in every gap (168)
- Cap-Tor xd color-matched ONLY at: first row 1 per joist on the house edge · borders 2 per bearing point @ 16" · fascia 2 at each board end then alternate top/bottom every 12", pre-drilled
- Rows per zone: A 24. Borders and breakers first, then the field rows from the outer edge toward the house.

CHECK

- 3/16" gaps held
- No cut end shows at the outer edge
- Color: every board from the same run



Step 7 · Rail — Fulton 36" Black

T-406 · Step 7 · Rail

GOES IN

- (4) TimberTech Fulton Rail 6' x 36" level panel Black
- (1) TimberTech Fulton Rail 8' x 36" level panel Black (pre-assembled steel balusters, cut to bay)
- (2) TimberTech Fulton 2" END post 36" Black w/ brackets, cap, skirt
- (2) TimberTech Fulton 2" LINE post 36" Black w/ brackets, cap, skirt
- (1) Rust-Oleum flat black touch-up (cut ends)

HOW

- Fulton 36" — 6 posts (2 CORNER, 2 END, 2 LINE); the 2 front corners are the shade-sail posts, base-plated on the deck — the panels bracket to them inside the outer rim ply, inner ply pocketed 2" wide, (2) 7/16" x 4-1/2" bolts per post

CHECK

- Top of rail 36" above the decking; posts plumb
- Cables tensioned / balusters under 4" clear
- End of construction

3D MODEL



The interactive 3D model lives at the online version of this document.

Prices: rate card · GC at cost · sold at $13,000 · 31.9% margin on the sold price (32.8% on the work) · financed ÷ 0.93 unless the job says otherwise. Internal — not for the homeowner.